
*India-Myanmar Center for Enhancement of IT Skills
Professional Diploma in AI
Natural Language Processing (Deep Learning)*

Assignment-I

Use 1D Convolutional Neural Networks (CNN) to classify text. You can use IMDB or SMS Spam Collection dataset. The model consists of an Embedding layer followed by a 1D convolutional layer (Conv1D), a MaxPooling1D layer, a Flatten layer, and fully connected Dense layers. The final layer produces the text classification result.

Embedding → Conv1D → MaxPooling1D → Flatten → Dense → Output

Questions

- (1) Compare different filter sizes:
 - a. kernel_size = 2
 - b. kernel_size = 3
 - c. kernel_size = 5
 - (2) Why do we use MaxPooling?
 - (3) What happens when the kernel size changes?
-

Assignment-II

Use a Transformer/BERT model for Named Entity Recognition (NER)

For example:

Barack Obama was born in Hawaii.

Expected output:

Barack Obama → PERSON

Hawaii → LOCATION

Use a pretrained BERT based NER model and experiment with sentences containing:

-PERSON
-ORGANIZATION
-LOCATION
-DATE
-MISC

You must create at least 20 test sentences.

Example:

- Apple released a new iPhone in California.
- Elon Musk founded SpaceX.
- Google has an office in Singapore.

For each sentence, display like this:

Entity	Label	Score
Microsoft	ORG	0.99
Bill Gates	PER	0.98
United States	LOC	0.97

Questions

- (1) Why is BERT called a contextualized language model?
 - (2) What is the difference between Word2Vec and BERT?
-

Assignment-III

Build an Encoder-Decoder LSTM that translate English sentences into French.

Example

English: I am happy.

French: Je suis heureux.

Architecture is

English → LSTM (encoder) → Context Vector → LSTM (decoder) → Dense → French

Questions

- (1) You must implement encoder and decoder and train using **epochs =50,100,150** and compare the results.
 - (2) You must experiment with **LSTM units = 128, 256, 512.**
 - (3) You must implement prediction
For example:
Input : Hello
Output: Bon jour.
 - (4) What are state_h and state_c?
 - (5) What is teacher forcing?
 - (6) What happens when the sentence becomes very long?
-

Assignment-IV

Questions

- (1) What is ASR?
- (2) Why can background noise reduce ASR performance?
- (3) Why is speech recognition harder than simple audio classification?

***** The End*****